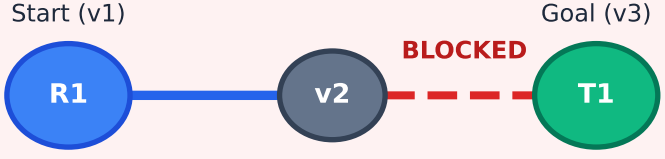
(a) Classical Auctions (CBBA / SSI): Cascading Reset

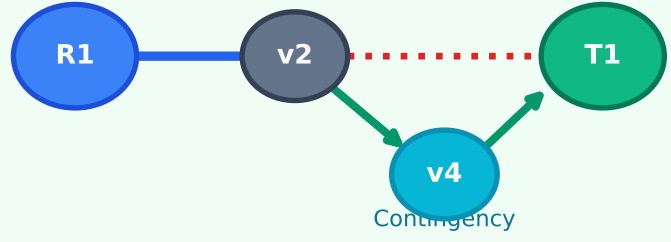


Failure Sequence:

1. R1 reaches blocked edge (v2 -> v3). Single path fails.
2. R1 drops Task T1 and broadcasts global nullification.
3. **Entire fleet forced into expensive multi-round re-auction.**
4. **Under jammed/partitioned comms: Allocation collapse.**

Result: 68% Completion, High Comm Surge (>75 msgs)

(b) Proposed ACA: Autonomous Contingency Tree Activation



ACA Resilience Sequence:

1. Precomputed Contingency Tree pre-calculates detour v4.
2. R1 detects blockage, updates P(theta), pivots locally to v4.
3. **Zero global re-auction messages sent; market stays intact.**

Result: 91% Completion, 0 Re-Auctions, Bounded Comms